Analysis of biological samples:
Technical summary of methods and quality assurance procedures
Prepared for the National Park Service and Montana Department of Environmental Quality
Andrew Ray, Project Manager
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by
W. Bollman, Chief Biologist
Rhithron Associates, Inc.
Missoula, Montana

METHODS

Sample processing

Seven macroinvertebrate samples from the 2018 Soda Butte Creek Project were delivered to Rhithron's laboratory facility in Missoula, Montana on March 23, 2019. A chain of custody document was provided by the Project Manager and was used as the inventory. Upon arrival, samples were unpacked and examined, and identifiers on each jar label were checked against the chain of custody. All samples arrived in good condition. An inventory spreadsheet was created which included project code and internal laboratory identification numbers and was uploaded into the Rhithron database prior to sample processing.

Subsamples of a minimum of 500 organisms were obtained using EMAP protocols (USEPA 2004) and Montana Department of Environmental Quality (MDEQ) standard procedures (MDEQ 2012): Caton sub-sampling devices (Caton 1991), divided into 30 grids, each approximately 6 cm by 6 cm were used. Each individual sample was thoroughly mixed in its jar(s), poured out and evenly spread into the Caton tray, and individual grids were randomly selected. Technicians assessed organism density in each sample prior to sorting in order to comply with the multiple MDEQ SOP requirements of a) a target number of 500 (± 10%) organisms and b) the need to completely pick the last selected grid. If organism density was high, technicians reduced the grid size and created a 120 grid matrix on the tray. If organism density was moderate, the entire Caton tray was divided into 30 grids. If the amount of detritus was too sparse to spread over the entire Caton tray, technicians evenly distributed it over a smaller portion of the tray and divided that portion into 30 appropriately sized grids. Once the sample was distributed appropriately individual grids were randomly selected. The contents of each grid were examined under stereoscopic microscopes using 10x-30x magnification. All aquatic invertebrates from each selected grid were sorted from the substrate, and placed in 80% ethanol for subsequent identification. Grid selection, examination, and sorting continued until at least 500 organisms were sorted. The final grid was completely sorted of all organisms.

Organisms were individually examined by certified taxonomists, using 10x – 80x stereoscopic dissecting scopes (Leica S8E) and identified to the lowest practical level consistent with MDEQ (MDEQ 2012) data requirements, using appropriate published taxonomic references and keys.

Identification, counts, life stages, and information about the condition of specimens were recorded on electronic bench sheets. Organisms that could not be identified to the taxonomic targets because of immaturity, poor condition, or lack of complete current regionally-applicable published keys were left at appropriate taxonomic levels that were coarser than those specified. To obtain accuracy in richness measures, these organisms were designated as "not unique" if other specimens from the same group could be taken to target levels. Organisms designated as "unique" were those that could be definitively distinguished from other organisms in the sample. Identified organisms were preserved in 80% ethanol in labeled vials, and archived at the Rhithron laboratory.

Chironomids and oligochaetes were carefully morphotyped using 10x – 80x stereoscopic dissecting microscopes (Leica S8E) and representative specimens were slide mounted and examined at 200x – 1000x magnification using an Olympus BX 51 or Leica DM 1000 compound microscope. Slide mounted organisms were archived at the Rhithron laboratory.

Quality control procedures

Internal quality control procedures for initial sample processing and subsampling involved checking sorting efficiency. These checks were conducted on a random selection of 10% of the samples by independent observers who microscopically re-examined 100% of sorted substrate from each sample. Quality control procedures for each sample proceeded as follows:

The quality control technician poured the sorted substrate from a processed sample out into a Caton tray, redistributing the substrate so that all of it could be accurately lifted out by removing entire grids in a random fashion. Grids were selected, and re-examined until 100% of the substrate was re-sorted. All organisms that were missed were counted and this number was added to the total number obtained in the original sort. Sorting efficiency was evaluated by applying the following calculation:

$$SE = \frac{n_1}{n_1 + n_2} \times 100$$

where: SE is the sorting efficiency, expressed as a percentage, n_1 is the total number of specimens in the first sort, and n_2 is the total number of specimens in the second sort.

Internal quality control procedures for taxonomic determinations of invertebrates involved checking accuracy, precision and enumeration. One sample was randomly selected and all organisms re-identified and counted by an independent taxonomist. Taxa lists and enumerations were compared by calculating a Bray-Curtis similarity statistic (Bray and Curtis 1957), Percent Taxonomic Disagreement (PTD) and Percent Difference in Enumeration (PDE). Routinely, discrepancies between the original identifications and the QC identifications are discussed among the taxonomists, and necessary rectifications to the data are made. Discrepancies that cannot be rectified by discussions are routinely sent out to taxonomic specialists for identification. However, taxonomic certainty for identifications in this project was high and no external verifications were necessary.

Data analysis

Taxa lists and counts for each sample were constructed. Standard metric calculations for aquatic invertebrate assemblages were made using Rhithron's customized laboratory information management system (LIMS). Calculations for the Montana DEQ HBI and O:E models were also performed (MDEQ 2012). A formatted data file for upload to the MT-eWQX database was generated.

RESULTS

Quality Control Procedures

Results of internal quality control procedures for subsampling and taxonomy are given in Table 1. Sorting efficiency was 98.73% for the randomly selected sort QC sample. Taxonomic precision for identification and enumeration measured by the Bray-Curtis index was 99.07%, PTD was 1.49% and PDE was 0.56% for the randomly selected taxonomic QC sample, and data entry efficiency was 100% for the project. These similarity statistics fall within acceptable industry criteria (Stribling et al. 2003).

Data analysis

Electronic spreadsheets were provided to the Project Manager via e-mail. Taxa lists are provided in an Appendix to this report.

REFERENCES

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Table 1. *Results of internal quality control procedures for subsampling and taxonomy.* Soda Butte Creek Project, 2018.

RAI Sample ID	Site ID	Site name	Collection date	Sorting efficiency	Bray-Curtis similarity for taxonomy and enumeration	Percent Taxonomic Disagreement (PTD)	Percent Difference in Enumeration (PDE)
MTDEQ18AR001	SBC-4	Soda Butte Creek upstream of the Yellowstone NP northeast boundary	9/21/2018				
MTDEQ18AR002	SBC-2	Soda Butte Creek downstream of the McLaren Mill and Tailings Reclamation site	9/21/2018	98.73%	99.07%	1.49%	0.56%
MTDEQ18AR003	SBC-RC	Soda Butte Creek downstream of the confluence of Republic Creek	9/22/2018				
MTDEQ18AR004	SBC-WY	Soda Butte Creek downstream of the confluence of Wyoming Creek	9/22/2018				
MTDEQ18AR005	MC-1 (East)	Miller Creek channel in reclamation site and upstream of confluence with Soda Butte Creek	9/23/2018				
MTDEQ18AR006	SBC-1	Soda Butte Creek upstream of the McLaren Mill and Tailings Reclamation site	9/23/2018				
MTDEQ18AR007	UT-1	Unnamed tributary upstream of confluence with Soda Butte Creek	9/23/2018				

APPENDIX

Taxa Lists

Soda Butte Creek
2018

Taxa Listing

Project ID: MTDEQ18AR
RAI No.: MTDEQ18AR001

RAI No.: **MTDEQ18AR001**
Client ID: **SBC-4**
Date Coll.: **9/21/2018** No. Jars: **2**
Sta. Name: **Soda Butte Creek upstream of the Yellowstone NP northeast boundary**
STORET ID:

Taxonomic Name	Count	PRA	Unique	Stage	Qualifier	BI	Function
Other Non-Insect							
Planariidae							
Polycelis sp.	**7**	**1.29%**	**Yes**	**Unknown**		**1**	**OM**
Nemata							
Nemata	**4**	**0.74%**	**Yes**	**Unknown**		**5**	**UN**
Acari							
Acari	**2**	**0.37%**	**Yes**	**Unknown**		**5**	**PR**
Ephemeroptera							
Ameletidae							
Ameletus sp.	**7**	**1.29%**	**Yes**	**Larva**		**0**	**SC**
Baetidae							
Acentrella sp.	**2**	**0.37%**	**Yes**	**Larva**		**4**	**CG**
Baetis bicaudatus complex	**4**	**0.74%**	**Yes**	**Larva**		**1**	**CG**
Baetis Rhodani Gr.	**2**	**0.37%**	**No**	**Larva**	**Early Instar**	**11**	**CG**
Baetis tricaudatus complex	**4**	**0.74%**	**Yes**	**Larva**		**5**	**CG**
Ephemerellidae							
Drunella sp.	**1**	**0.18%**	**Yes**	**Larva**	**Early Instar**	**1**	**SC**
Drunella coloradensis	**1**	**0.18%**	**Yes**	**Larva**		**0**	**SC**
Drunella doddsii	**14**	**2.58%**	**Yes**	**Larva**		**1**	**SC**
Ephemerella sp.	**15**	**2.76%**	**Yes**	**Larva**	**Early Instar**	**1.5**	**SC**
Heptageniidae							
Cinygmula sp.	**47**	**8.66%**	**Yes**	**Larva**		**0**	**SC**
Epeorus deceptivus	**1**	**0.18%**	**Yes**	**Larva**		**0**	**SC**
Epeorus grandis	**3**	**0.55%**	**Yes**	**Larva**		**0**	**SC**
Rhithrogena sp.	**13**	**2.39%**	**Yes**	**Larva**		**0**	**SC**
Plecoptera							
Chloroperlidae							
Chloroperlidae	**7**	**1.29%**	**No**	**Larva**	**Early Instar**	**1**	**PR**
Sweltsa sp.	**1**	**0.18%**	**Yes**	**Larva**		**0**	**PR**
Nemouridae							
Zapada cinctipes	**1**	**0.18%**	**Yes**	**Larva**		**3**	**SH**
Zapada columbiana	**10**	**1.84%**	**Yes**	**Larva**		**2**	**SH**
Perlodidae							
Megarcys sp.	**7**	**1.29%**	**Yes**	**Larva**		**1**	**PR**
Taeniopterygidae							
Taeniopterygidae	**19**	**3.50%**	**Yes**	**Larva**	**Early Instar**	**2**	**SH**
Trichoptera							
Glossosomatidae							
Glossosoma sp.	**28**	**5.16%**	**Yes**	**Larva**		**0**	**SC**
Hydropsychidae							
Hydropsychidae	**8**	**1.47%**	**Yes**	**Larva**	**Early Instar**	**4**	**CF**
Limnephilidae							
Limnephilidae	**1**	**0.18%**	**Yes**	**Larva**	**Early Instar**	**3**	**SH**
Rhyacophilidae							
Rhyacophila atrata complex	**2**	**0.37%**	**Yes**	**Larva**		**0**	**PR**
Rhyacophila Hyalinata Gr.	**1**	**0.18%**	**Yes**	**Larva**		**0**	**PR**
Rhyacophila Vofixa Gr.	**6**	**1.10%**	**Yes**	**Larva**		**0**	**PR**

Taxa Listing

Project ID: MTDEQ18AR
RAI No.: MTDEQ18AR001

RAI No.: **MTDEQ18AR001**
Client ID: **SBC-4**
Date Coll.: **9/21/2018**
No. Jars: **2**
Sta. Name: **Soda Butte Creek upstream of the Yellowstone NP northeast boundary**
STORET ID:

Taxonomic Name	Count	PRA	Unique	Stage	Qualifier	BI	Function
Diptera							
Ceratopogonidae							
Ceratopogoninae	**12**	**2.21%**	**Yes**	**Larva**		**6**	**PR**
Psychodidae							
Pericoma sp.	**43**	**7.92%**	**Yes**	**Larva**		**4**	**CG**
Simuliidae							
Simulium sp.	**1**	**0.18%**	**Yes**	**Larva**		**6**	**CF**
Chironomidae							
Chironominae							
Micropsectra sp.	**1**	**0.18%**	**Yes**	**Larva**		**4**	**CG**
Tanytarsini	**4**	**0.74%**	**No**	**Larva**	**Early Instar**	**6**	**CF**
Diamesinae							
Diamesa sp.	**7**	**1.29%**	**Yes**	**Larva**		**5**	**CG**
Diamesa sp.	**3**	**0.55%**	**No**	**Pupa**		**5**	**CG**
Pagastia sp.	**2**	**0.37%**	**Yes**	**Larva**		**1**	**CG**
Potthastia sp.	**1**	**0.18%**	**No**	**Pupa**		**2**	**CG**
Potthastia Gaedii Gr.	**3**	**0.55%**	**Yes**	**Larva**		**2**	**CG**
Orthocladiinae							
Cricotopus sp.	**1**	**0.18%**	**No**	**Pupa**		**7**	**SH**
Cricotopus (Nostococladius) sp.	**1**	**0.18%**	**Yes**	**Larva**		**6**	**SH**
Cricotopus / Orthocladius	**62**	**11.42%**	**No**	**Larva**	**Early Instar**	**7**	**SH**
Eukiefferiella Devonica Gr.	**1**	**0.18%**	**Yes**	**Larva**		**8**	**CG**
Eukiefferiella Gracei Gr.	**5**	**0.92%**	**Yes**	**Larva**		**8**	**CG**
Orthocladiinae	**1**	**0.18%**	**No**	**Pupa**	**Damaged**	**6**	**CG**
Orthocladius sp.	**135**	**24.86%**	**Yes**	**Larva**		**6**	**CG**
Orthocladius sp.	**34**	**6.26%**	**No**	**Pupa**		**6**	**CG**
Rheocricotopus sp.	**4**	**0.74%**	**Yes**	**Larva**		**4**	**CG**
Thienemanniella sp.	**1**	**0.18%**	**Yes**	**Larva**		**6**	**CG**
Tvetenia Bavarica Gr.	**3**	**0.55%**	**Yes**	**Larva**		**5**	**CG**
Sample Count	543						

Taxa Listing

Project ID: MTDEQ18AR
RAI No.: MTDEQ18AR002

RAI No.: **MTDEQ18AR002**
Client ID: **SBC-2**
Date Coll.: **9/21/2018** No. Jars: **2**
Sta. Name: **Soda Butte Creek downstream of the McLaren Mill and Tailings Reclamation site**
STORET ID:

Taxonomic Name	Count	PRA	Unique	Stage	Qualifier	BI	Function
Other Non-Insect							
Planariidae							
Polycelis sp.	52	9.67%	Yes	Unknown		1	OM
Nemata							
Nemata	1	0.19%	Yes	Unknown		5	UN
Ostracoda							
Ostracoda	8	1.49%	Yes	Unknown		8	CG
Acari							
Acari	16	2.97%	Yes	Unknown		5	PR
Ephemeroptera							
Baetidae							
Baetis Rhodani Gr.	7	1.30%	Yes	Larva	Early Instar	11	CG
Ephemerellidae							
Drunella coloradensis	2	0.37%	Yes	Larva		0	SC
Drunella doddsii	2	0.37%	Yes	Larva		1	SC
Ephemerella sp.	34	6.32%	Yes	Larva	Early Instar	1.5	SC
Heptageniidae							
Cinygmula sp.	28	5.20%	Yes	Larva		0	SC
Epeorus grandis	12	2.23%	Yes	Larva		0	SC
Rhithrogena sp.	4	0.74%	Yes	Larva		0	SC
Plecoptera							
Chloroperlidae							
Paraperla sp.	2	0.37%	Yes	Larva		1	CG
Sweltsa sp.	13	2.42%	Yes	Larva		0	PR
Nemouridae							
Zapada cinctipes	3	0.56%	Yes	Larva		3	SH
Zapada columbiana	64	11.90%	Yes	Larva		2	SH
Perlodidae							
Megarcys sp.	8	1.49%	Yes	Larva		1	PR
Taeniopterygidae							
Taeniopterygidae	11	2.04%	Yes	Larva	Early Instar	2	SH
Trichoptera							
Glossosomatidae							
Glossosoma sp.	3	0.56%	Yes	Larva		0	SC
Hydropsychidae							
Parapsyche elsis	48	8.92%	Yes	Larva		1	PR
Limnephilidae							
Limnephilidae	2	0.37%	Yes	Larva	Early Instar	3	SH
Rhyacophilidae							
Rhyacophila sp.	12	2.23%	No	Larva	Early Instar	1	PR
Rhyacophila atrata complex	5	0.93%	Yes	Larva		0	PR
Rhyacophila Betteni Gr.	1	0.19%	Yes	Larva		0	PR
Rhyacophila Brunnea/Vemna Gr.	1	0.19%	Yes	Larva		2	PR
Rhyacophila Hyalinata Gr.	11	2.04%	Yes	Larva		0	PR
Rhyacophila Vofixa Gr.	22	4.09%	Yes	Larva		0	PR
Thremmatidae							
Thremmatidae	4	0.74%	Yes	Larva	Early Instar	11	SC

Taxa Listing

Project ID: MTDEQ18AR
RAI No.: MTDEQ18AR002

RAI No.: **MTDEQ18AR002**
Client ID: **SBC-2**
Date Coll.: **9/21/2018**
No. Jars: **2**
Sta. Name: **Soda Butte Creek downstream of the McLaren Mill and Tailings Reclamation site**
STORET ID:

Taxonomic Name	Count	PRA	Unique	Stage	Qualifier	BI	Function
Diptera							
Ceratopogonidae							
Ceratopogoninae	**54**	**10.04%**	**Yes**	**Larva**		**6**	**PR**
Empididae							
Empididae	**1**	**0.19%**	**Yes**	**Pupa**		**6**	**PR**
Psychodidae							
Pericoma sp.	**18**	**3.35%**	**Yes**	**Larva**		**4**	**CG**
Tipulidae							
Dicranota sp.	**1**	**0.19%**	**Yes**	**Larva**		**3**	**PR**
Chironomidae							
Chironominae							
Polypedilum sp.	**1**	**0.19%**	**Yes**	**Larva**		**6**	**SH**
Chironominae							
Micropsectra sp.	**2**	**0.37%**	**Yes**	**Larva**		**4**	**CG**
Diamesinae							
Diamesa sp.	**1**	**0.19%**	**Yes**	**Pupa**		**5**	**CG**
Pagastia sp.	**2**	**0.37%**	**Yes**	**Larva**		**1**	**CG**
Orthocladiinae							
Brillia sp.	**11**	**2.04%**	**Yes**	**Larva**		**4**	**SH**
Cricotopus / Orthocladius	**35**	**6.51%**	**No**	**Larva**	**Early Instar**	**7**	**SH**
Orthocladius sp.	**21**	**3.90%**	**Yes**	**Larva**		**6**	**CG**
Orthocladius sp.	**1**	**0.19%**	**No**	**Pupa**		**6**	**CG**
Parorthocladius sp.	**3**	**0.56%**	**Yes**	**Larva**		**6**	**CG**
Parorthocladius sp.	**1**	**0.19%**	**No**	**Pupa**		**6**	**CG**
Rheocricotopus sp.	**1**	**0.19%**	**Yes**	**Larva**		**4**	**CG**
Stilocladius sp.	**2**	**0.37%**	**Yes**	**Larva**		**3**	**CG**
Thienemanniella sp.	**1**	**0.19%**	**Yes**	**Larva**		**6**	**CG**
Tvetenia Bavarica Gr.	**6**	**1.12%**	**Yes**	**Larva**		**5**	**CG**
Sample Count	538						

Taxa Listing

Project ID: MTDEQ18AR
RAI No.: MTDEQ18AR003

RAI No.: **MTDEQ18AR003**
Client ID: **SBC-RC**
Date Coll.: **9/22/2018**
No. Jars: **2**
Sta. Name: **Soda Butte Creek downstream of the confluence of Republic Creek**
STORET ID:

Taxonomic Name	Count	PRA	Unique	Stage	Qualifier	BI	Function
Other Non-Insect							
Planariidae							
Polycelis sp.	8	1.49%	Yes	Unknown		1	OM
Nemata							
Nemata	2	0.37%	Yes	Unknown		5	UN
Ostracoda							
Ostracoda	1	0.19%	Yes	Unknown		8	CG
Acari							
Acari	8	1.49%	Yes	Unknown		5	PR
Ephemeroptera							
Ameletidae							
Ameletus sp.	74	13.81%	Yes	Larva		0	SC
Baetidae							
Baetis tricaudatus complex	6	1.12%	Yes	Larva		5	CG
Ephemerellidae							
Drunella doddsii	18	3.36%	Yes	Larva		1	SC
Ephemerella sp.	4	0.75%	Yes	Larva	Early Instar	1.5	SC
Heptageniidae							
Cinygmula sp.	71	13.25%	Yes	Larva		0	SC
Epeorus deceptivus	9	1.68%	Yes	Larva		0	SC
Epeorus grandis	44	8.21%	Yes	Larva		0	SC
Rhithrogena sp.	55	10.26%	Yes	Larva		0	SC
Plecoptera							
Chloroperlidae							
Paraperla sp.	25	4.66%	Yes	Larva		1	CG
Nemouridae							
Zapada columbiana	70	13.06%	Yes	Larva		2	SH
Perlidae							
Doroneuria sp.	1	0.19%	Yes	Larva		0	PR
Perlodidae							
Megarcys sp.	10	1.87%	Yes	Larva		1	PR
Taeniopterygidae							
Taeniopterygidae	8	1.49%	Yes	Larva	Early Instar	2	SH
Trichoptera							
Apataniidae							
Apatania sp.	3	0.56%	Yes	Larva		3	SC
Glossosomatidae							
Glossosoma sp.	5	0.93%	Yes	Larva		0	SC
Hydropsychidae							
Parapsyche elsis	28	5.22%	Yes	Larva		1	PR
Limnephilidae							
Limnephilidae	1	0.19%	Yes	Larva	Early Instar	3	SH
Rhyacophilidae							
Rhyacophila sp.	1	0.19%	No	Pupa		1	PR
Rhyacophila Hyalinata Gr.	7	1.31%	Yes	Larva		0	PR
Rhyacophila Vofixa Gr.	30	5.60%	Yes	Larva		0	PR

Taxa Listing

Project ID: MTDEQ18AR
RAI No.: MTDEQ18AR003

RAI No.: **MTDEQ18AR003**
Client ID: **SBC-RC**
Date Coll.: **9/22/2018** No. Jars: **2**
Sta. Name: **Soda Butte Creek downstream of the confluence of Republic Creek**
STORET ID:

Taxonomic Name	Count	PRA	Unique	Stage	Qualifier	BI	Function
Diptera							
Ceratopogonidae							
Ceratopogoninae	**14**	**2.61%**	**Yes**	**Larva**		**6**	**PR**
Empididae							
Chelifera sp.	**1**	**0.19%**	**Yes**	**Larva**		**5**	**PR**
Psychodidae							
Pericoma sp.	**7**	**1.31%**	**Yes**	**Larva**		**4**	**CG**
Tipulidae							
Rhabdomastix Setigera Gr.	**1**	**0.19%**	**Yes**	**Larva**		**3**	**CG**
Chironomidae							
Chironominae							
Micropsectra sp.	**1**	**0.19%**	**Yes**	**Larva**		**4**	**CG**
Diamesinae							
Diamesa sp.	**1**	**0.19%**	**Yes**	**Larva**		**5**	**CG**
Orthocladiinae							
Brillia sp.	**1**	**0.19%**	**Yes**	**Larva**		**4**	**SH**
Chaetocladius sp.	**4**	**0.75%**	**Yes**	**Larva**		**6**	**CG**
Cricotopus (Nostococladius) sp.	**2**	**0.37%**	**Yes**	**Larva**		**6**	**SH**
Orthocladiinae	**1**	**0.19%**	**No**	**Larva**	**Damaged**	**6**	**CG**
Orthocladius sp.	**2**	**0.37%**	**Yes**	**Larva**		**6**	**CG**
Rheocricotopus sp.	**10**	**1.87%**	**Yes**	**Larva**		**4**	**CG**
Tvetenia Bavarica Gr.	**2**	**0.37%**	**Yes**	**Larva**		**5**	**CG**
Sample Count	536						

Taxa Listing

Project ID: MTDEQ18AR
RAI No.: MTDEQ18AR004

RAI No.: **MTDEQ18AR004**
Client ID: **SBC-WY**
Date Coll.: **9/22/2018** No. Jars: **2**
Sta. Name: **Soda Butte Creek downstream of the confluence of Wyoming Creek**
STORET ID:

Taxonomic Name	Count	PRA	Unique	Stage	Qualifier	BI	Function
Other Non-Insect							
Nemata							
Nemata	**2**	**0.37%**	**Yes**	**Unknown**		**5**	**UN**
Acari							
Acari	**38**	**6.95%**	**Yes**	**Unknown**		**5**	**PR**
Ephemeroptera							
Ameletidae							
Ameletus sp.	**22**	**4.02%**	**Yes**	**Larva**		**0**	**SC**
Baetidae							
Baetis tricaudatus complex	**27**	**4.94%**	**Yes**	**Larva**		**5**	**CG**
Ephemerellidae							
Drunella coloradensis	**1**	**0.18%**	**Yes**	**Larva**		**0**	**SC**
Drunella doddsii	**27**	**4.94%**	**Yes**	**Larva**		**1**	**SC**
Ephemerella sp.	**1**	**0.18%**	**Yes**	**Larva**	**Early Instar**	**1.5**	**SC**
Heptageniidae							
Cinygmula sp.	**7**	**1.28%**	**Yes**	**Larva**		**0**	**SC**
Epeorus grandis	**21**	**3.84%**	**Yes**	**Larva**		**0**	**SC**
Rhithrogena sp.	**3**	**0.55%**	**Yes**	**Larva**		**0**	**SC**
Plecoptera							
Chloroperlidae							
Paraperla sp.	**7**	**1.28%**	**Yes**	**Larva**		**1**	**CG**
Nemouridae							
Zapada columbiana	**90**	**16.45%**	**Yes**	**Larva**		**2**	**SH**
Perlodidae							
Megarcys sp.	**4**	**0.73%**	**Yes**	**Larva**		**1**	**PR**
Taeniopterygidae							
Taeniopterygidae	**8**	**1.46%**	**Yes**	**Larva**	**Early Instar**	**2**	**SH**
Trichoptera							
Glossosomatidae							
Glossosoma sp.	**1**	**0.18%**	**Yes**	**Larva**		**0**	**SC**
Glossosomatidae	**10**	**1.83%**	**No**	**Pupa**		**0**	**SC**
Hydropsychidae							
Arctopsyche sp.	**1**	**0.18%**	**Yes**	**Larva**		**2**	**PR**
Parapsyche elsis	**21**	**3.84%**	**Yes**	**Larva**		**1**	**PR**
Limnephilidae							
Dicosmoecus atripes	**1**	**0.18%**	**Yes**	**Larva**		**1**	**SC**
Rhyacophilidae							
Rhyacophila Hyalinata Gr.	**8**	**1.46%**	**Yes**	**Larva**		**0**	**PR**
Rhyacophila Vofixa Gr.	**82**	**14.99%**	**Yes**	**Larva**		**0**	**PR**

Taxa Listing

Project ID: MTDEQ18AR
RAI No.: MTDEQ18AR004

RAI No.: **MTDEQ18AR004**
Client ID: **SBC-WY**
Date Coll.: **9/22/2018**
No. Jars: **2**
Sta. Name: **Soda Butte Creek downstream of the confluence of Wyoming Creek**
STORET ID:

Taxonomic Name	Count	PRA	Unique	Stage	Qualifier	BI	Function
Diptera							
Ceratopogonidae							
Ceratopogoninae	**21**	**3.84%**	**Yes**	**Larva**		**6**	**PR**
Psychodidae							
Pericoma sp.	**75**	**13.71%**	**Yes**	**Larva**		**4**	**CG**
Simuliidae							
Simulium sp.	**1**	**0.18%**	**Yes**	**Larva**		**6**	**CF**
Tipulidae							
Rhabdomastix Fascigera Gr.	**1**	**0.18%**	**Yes**	**Larva**		**1**	**PR**
Chironomidae							
Diamesinae							
Diamesa sp.	**4**	**0.73%**	**Yes**	**Larva**		**5**	**CG**
Pagastia sp.	**3**	**0.55%**	**Yes**	**Larva**		**1**	**CG**
Orthocladiinae							
Cricotopus / Orthocladius	**22**	**4.02%**	**No**	**Larva**	**Early Instar**	**7**	**SH**
Eukiefferiella Gracei Gr.	**3**	**0.55%**	**Yes**	**Larva**		**8**	**CG**
Orthocladiinae	**1**	**0.18%**	**No**	**Pupa**	**Damaged**	**6**	**CG**
Orthocladiinae	**2**	**0.37%**	**No**	**Larva**	**Early Instar**	**6**	**CG**
Orthocladius sp.	**30**	**5.48%**	**Yes**	**Larva**		**6**	**CG**
Tvetenia Bavarica Gr.	**2**	**0.37%**	**Yes**	**Larva**		**5**	**CG**
Sample Count	547						

Taxa Listing

Project ID: MTDEQ18AR
RAI No.: MTDEQ18AR005

RAI No.: **MTDEQ18AR005** Sta. Name: **Miller Creek channel in reclamation site and upstream of confluence with Soda Butte Creek**
Client ID: **MC-1 (East)**
Date Coll.: **9/23/2018** No. Jars: **3** STORET ID:

Taxonomic Name	Count	PRA	Unique	Stage	Qualifier	BI	Function
Other Non-Insect							
Planariidae							
Polycelis sp.	65	14.04%	Yes	Unknown		1	OM
Enchytraeidae							
Enchytraeus sp.	3	0.65%	Yes	Unknown		4	CG
Fridericia sp.	6	1.30%	Yes	Unknown		11	CG
Ostracoda							
Ostracoda	7	1.51%	Yes	Unknown		8	CG
Acari							
Acari	8	1.73%	Yes	Unknown		5	PR
Ephemeroptera							
Ameletidae							
Ameletus sp.	5	1.08%	Yes	Larva		0	SC
Baetidae							
Baetis tricaudatus complex	1	0.22%	Yes	Larva		5	CG
Ephemerellidae							
Ephemerella sp.	6	1.30%	Yes	Larva	Early Instar	1.5	SC
Heptageniidae							
Cinygmula sp.	60	12.96%	Yes	Larva		0	SC
Epeorus grandis	21	4.54%	Yes	Larva		0	SC
Heptageniidae	4	0.86%	No	Larva	Early Instar	4	SC
Rhithrogena sp.	67	14.47%	Yes	Larva		0	SC
Plecoptera							
Capniidae							
Capniidae	18	3.89%	Yes	Larva	Early Instar	1	SH
Chloroperlidae							
Chloroperlidae	9	1.94%	No	Larva	Early Instar	1	PR
Paraperla sp.	36	7.78%	Yes	Larva		1	CG
Sweltsa sp.	8	1.73%	Yes	Larva		0	PR
Nemouridae							
Visoka cataractae	1	0.22%	Yes	Larva		0	SH
Zapada columbiana	14	3.02%	Yes	Larva		2	SH
Zapada Oregonensis Gr.	1	0.22%	Yes	Larva		2	SH
Perlodidae							
Megarcys sp.	25	5.40%	Yes	Larva		1	PR
Trichoptera							
Hydropsychidae							
Parapsyche elsis	27	5.83%	Yes	Larva		1	PR
Rhyacophilidae							
Rhyacophila sp.	4	0.86%	No	Larva	Early Instar	1	PR
Rhyacophila atrata complex	3	0.65%	Yes	Larva		0	PR
Rhyacophila Betteni Gr.	26	5.62%	Yes	Larva		0	PR
Rhyacophila Brunnea/Vemna Gr.	1	0.22%	Yes	Larva		2	PR
Rhyacophila Hyalinata Gr.	10	2.16%	Yes	Larva		0	PR
Rhyacophila Vofixa Gr.	6	1.30%	Yes	Larva		0	PR
Thremmatidae							
Thremmatidae	2	0.43%	Yes	Larva	Early Instar	11	SC

Taxa Listing

Project ID: MTDEQ18AR
RAI No.: MTDEQ18AR005

RAI No.: **MTDEQ18AR005**
Client ID: **MC-1 (East)**
Date Coll.: **9/23/2018** No. Jars: **3**

Sta. Name: **Miller Creek channel in reclamation site and upstream of confluence with Soda Butte Creek**
STORET ID:

Taxonomic Name	Count	PRA	Unique	Stage	Qualifier	BI	Function
Diptera							
Ceratopogonidae							
Ceratopogoninae	**2**	**0.43%**	**Yes**	**Larva**		**6**	**PR**
Tipulidae							
Dicranota sp.	**1**	**0.22%**	**Yes**	**Larva**		**3**	**PR**
Chironomidae							
Diamesinae							
Diamesa sp.	**2**	**0.43%**	**Yes**	**Pupa**		**5**	**CG**
Pagastia sp.	**4**	**0.86%**	**Yes**	**Larva**		**1**	**CG**
Orthocladiinae							
Brillia sp.	**2**	**0.43%**	**Yes**	**Larva**		**4**	**SH**
Chaetocladius sp.	**1**	**0.22%**	**Yes**	**Larva**		**6**	**CG**
Cricotopus (Nostococladius) sp.	**5**	**1.08%**	**Yes**	**Larva**		**6**	**SH**
Cricotopus / Orthocladius	**1**	**0.22%**	**Yes**	**Larva**	**Early Instar**	**7**	**SH**
Eukiefferiella tirolensis	**1**	**0.22%**	**Yes**	**Larva**		**8**	**CG**
Sample Count	463						

Taxa Listing

Project ID: MTDEQ18AR
RAI No.: MTDEQ18AR006

RAI No.: **MTDEQ18AR006**
Client ID: **SBC-1**
Date Coll.: **9/23/2018**
No. Jars: **3**
Sta. Name: **Soda Butte Creek upstream of the McLaren Mill and Tailings Reclamation site**
STORET ID:

Taxonomic Name	Count	PRA	Unique	Stage	Qualifier	BI	Function
Other Non-Insect							
Planariidae							
Polycelis sp.	48	8.73%	Yes	Unknown		1	OM
Nemata							
Nemata	6	1.09%	Yes	Unknown		5	UN
Enchytraeidae							
Chamaedrilus sp.	1	0.18%	Yes	Unknown		11	UN
Henlea sp.	3	0.55%	Yes	Unknown		11	UN
Ostracoda							
Ostracoda	21	3.82%	Yes	Unknown		8	CG
Acari							
Acari	3	0.55%	Yes	Unknown		5	PR
Ephemeroptera							
Ameletidae							
Ameletus sp.	4	0.73%	Yes	Larva		0	SC
Baetidae							
Baetis Rhodani Gr.	11	2.00%	No	Larva	Damaged	11	CG
Baetis tricaudatus complex	5	0.91%	Yes	Larva		5	CG
Ephemerellidae							
Drunella sp.	1	0.18%	Yes	Larva	Early Instar	1	SC
Drunella coloradensis	9	1.64%	Yes	Larva		0	SC
Drunella doddsii	18	3.27%	Yes	Larva		1	SC
Ephemerella sp.	4	0.73%	Yes	Larva	Early Instar	1.5	SC
Heptageniidae							
Cinygmula sp.	7	1.27%	Yes	Larva		0	SC
Epeorus deceptivus	2	0.36%	Yes	Larva		0	SC
Epeorus grandis	1	0.18%	Yes	Larva		0	SC
Rhithrogena sp.	3	0.55%	Yes	Larva		0	SC
Plecoptera							
Chloroperlidae							
Chloroperlidae	16	2.91%	No	Larva	Early Instar	1	PR
Paraperla sp.	4	0.73%	Yes	Larva		1	CG
Sweltsa sp.	34	6.18%	Yes	Larva		0	PR
Leuctridae							
Leuctridae	2	0.36%	Yes	Larva	Early Instar	0	SH
Nemouridae							
Zapada columbiana	73	13.27%	Yes	Larva		2	SH
Perlodidae							
Isoperla sp.	14	2.55%	Yes	Larva		2	PR
Kogotus sp.	1	0.18%	Yes	Larva		1	PR
Megarcys sp.	4	0.73%	Yes	Larva		1	PR
Taeniopterygidae							
Taeniopterygidae	10	1.82%	Yes	Larva	Early Instar	2	SH

Taxa Listing

Project ID: MTDEQ18AR
RAI No.: MTDEQ18AR006

RAI No.: **MTDEQ18AR006**
Client ID: **SBC-1**
Date Coll.: **9/23/2018** No. Jars: **3**
Sta. Name: **Soda Butte Creek upstream of the McLaren Mill and Tailings Reclamation site**
STORET ID:

Taxonomic Name	Count	PRA	Unique	Stage	Qualifier	BI	Function
Trichoptera							
Glossosomatidae							
Glossosoma sp.	3	0.55%	Yes	Larva		0	SC
Hydropsychidae							
Parapsyche elsis	24	4.36%	Yes	Larva		1	PR
Limnephilidae							
Limnephilidae	1	0.18%	Yes	Larva	Early Instar	3	SH
Rhyacophilidae							
Rhyacophila sp.	2	0.36%	No	Larva	Early Instar	1	PR
Rhyacophila atrata complex	2	0.36%	Yes	Larva		0	PR
Rhyacophila Betteni Gr.	7	1.27%	Yes	Larva		0	PR
Rhyacophila Brunnea/Vemna Gr.	5	0.91%	Yes	Larva		2	PR
Rhyacophila Hyalinata Gr.	15	2.73%	Yes	Larva		0	PR
Rhyacophila narvae	13	2.36%	Yes	Larva		0	PR
Rhyacophila Vofixa Gr.	5	0.91%	Yes	Larva		0	PR
Diptera							
Ceratopogonidae							
Ceratopogoninae	3	0.55%	Yes	Larva		6	PR
Empididae							
Empididae	1	0.18%	No	Pupa		6	PR
Neoplasta sp.	2	0.36%	Yes	Larva		5	PR
Psychodidae							
Pericoma sp.	4	0.73%	Yes	Larva		4	CG
Simuliidae							
Prosimulium sp.	1	0.18%	Yes	Larva		4	CF
Simulium sp.	7	1.27%	Yes	Larva		6	CF
Chironomidae							
Diamesinae							
Diamesa sp.	1	0.18%	No	Pupa		5	CG
Diamesa sp.	2	0.36%	Yes	Larva		5	CG
Pagastia sp.	45	8.18%	Yes	Larva		1	CG
Orthocladiinae							
Brillia sp.	1	0.18%	Yes	Larva		4	SH
Chaetocladius sp.	3	0.55%	Yes	Larva		6	CG
Cricotopus / Orthocladius	14	2.55%	No	Larva	Early Instar	7	SH
Diplocladius cultriger	1	0.18%	Yes	Pupa		8	CG
Eukiefferiella sp.	3	0.55%	No	Pupa		8	CG
Eukiefferiella sp.	3	0.55%	No	Larva	Early Instar	8	CG
Eukiefferiella Claripennis Gr.	3	0.55%	Yes	Larva		8	CG
Eukiefferiella Gracei Gr.	11	2.00%	Yes	Larva		8	CG
Hydrobaenus sp.	2	0.36%	Yes	Larva		8	SC
Orthocladiinae	1	0.18%	No	Pupa	Damaged	6	CG
Orthocladiinae	1	0.18%	No	Larva	Early Instar	6	CG
Orthocladius sp.	54	9.82%	Yes	Larva		6	CG
Tvetenia sp.	1	0.18%	No	Pupa		5	CG
Tvetenia Bavarica Gr.	4	0.73%	Yes	Larva		5	CG

Taxa Listing

Project ID: MTDEQ18AR
RAI No.: MTDEQ18AR006

RAI No.: **MTDEQ18AR006**
Client ID: **SBC-1**
Date Coll.: **9/23/2018** No. Jars: **3**
Sta. Name: **Soda Butte Creek upstream of the McLaren Mill and Tailings Reclamation site**
STORET ID:

Taxonomic Name	Count	PRA	Unique	Stage	Qualifier	BI	Function
Sample Count	550						

Taxa Listing

Project ID: MTDEQ18AR
RAI No.: MTDEQ18AR007

RAI No.: **MTDEQ18AR007**
Client ID: **UT-1**
Date Coll.: **9/23/2018** No. Jars: **3**

Sta. Name: **Unnamed tributary upstream of confluence with Soda Butte Creek**
STORET ID:

Taxonomic Name	Count	PRA	Unique	Stage	Qualifier	BI	Function
Other Non-Insect							
Planariidae							
Polycelis sp.	**1**	**0.18%**	**Yes**	**Unknown**		**1**	**OM**
Nemata							
Nemata	**3**	**0.55%**	**Yes**	**Unknown**		**5**	**UN**
Enchytraeidae							
Mesenchytraeus sp.	**14**	**2.55%**	**Yes**	**Unknown**		**4**	**CG**
Acari							
Acari	**3**	**0.55%**	**Yes**	**Unknown**		**5**	**PR**
Ephemeroptera							
Ameletidae							
Ameletus sp.	**9**	**1.64%**	**Yes**	**Larva**		**0**	**SC**
Baetidae							
Baetis bicaudatus complex	**1**	**0.18%**	**Yes**	**Larva**		**1**	**CG**
Baetis Rhodani Gr.	**7**	**1.27%**	**No**	**Larva**	**Damaged**	**11**	**CG**
Baetis tricaudatus complex	**4**	**0.73%**	**Yes**	**Larva**		**5**	**CG**
Heptageniidae							
Cinygmula sp.	**8**	**1.45%**	**Yes**	**Larva**		**0**	**SC**
Epeorus deceptivus	**1**	**0.18%**	**Yes**	**Larva**		**0**	**SC**
Epeorus grandis	**75**	**13.64%**	**Yes**	**Larva**		**0**	**SC**
Rhithrogena sp.	**29**	**5.27%**	**Yes**	**Larva**		**0**	**SC**
Plecoptera							
Capniidae							
Capniidae	**1**	**0.18%**	**Yes**	**Larva**	**Early Instar**	**1**	**SH**
Chloroperlidae							
Paraperla sp.	**23**	**4.18%**	**Yes**	**Larva**		**1**	**CG**
Sweltsa sp.	**1**	**0.18%**	**Yes**	**Larva**		**0**	**PR**
Nemouridae							
Zapada sp.	**1**	**0.18%**	**No**	**Larva**	**Damaged**	**2**	**SH**
Zapada columbiana	**94**	**17.09%**	**Yes**	**Larva**		**2**	**SH**
Zapada Oregonensis Gr.	**87**	**15.82%**	**Yes**	**Larva**		**2**	**SH**
Perlodidae							
Diura sp.	**3**	**0.55%**	**Yes**	**Larva**		**2**	**PR**
Isoperla sp.	**14**	**2.55%**	**Yes**	**Larva**		**2**	**PR**
Megarcys sp.	**27**	**4.91%**	**Yes**	**Larva**		**1**	**PR**
Taeniopterygidae							
Taeniopterygidae	**11**	**2.00%**	**Yes**	**Larva**	**Early Instar**	**2**	**SH**

Taxa Listing

Project ID: MTDEQ18AR
RAI No.: MTDEQ18AR007

RAI No.: **MTDEQ18AR007**
Client ID: **UT-1**
Date Coll.: **9/23/2018** No. Jars: **3**
Sta. Name: **Unnamed tributary upstream of confluence with Soda Butte Creek**
STORET ID:

Taxonomic Name	Count	PRA	Unique	Stage	Qualifier	BI	Function
Trichoptera							
Hydropsychidae							
Parapsyche elsis	7	1.27%	Yes	Larva		1	PR
Limnephilidae							
Ecclisomyia sp.	2	0.36%	Yes	Larva		4	CG
Homophylax sp.	1	0.18%	Yes	Larva		2	SH
Limnephilidae	7	1.27%	No	Larva	Early Instar	3	SH
Rhyacophilidae							
Rhyacophila sp.	2	0.36%	No	Pupa		1	PR
Rhyacophila sp.	15	2.73%	No	Larva	Early Instar	1	PR
Rhyacophila Brunnea/Vemna Gr.	1	0.18%	Yes	Larva		2	PR
Rhyacophila Hyalinata Gr.	1	0.18%	Yes	Larva		0	PR
Rhyacophila Vofixa Gr.	1	0.18%	Yes	Larva		0	PR
Diptera							
Empididae							
Empididae	1	0.18%	Yes	Pupa		6	PR
Simuliidae							
Simuliidae	2	0.36%	Yes	Larva	Early Instar	6	CF
Tipulidae							
Dicranota sp.	6	1.09%	Yes	Larva		3	PR
Chironomidae							
Diamesinae							
Diamesa sp.	3	0.55%	Yes	Larva		5	CG
Pagastia sp.	11	2.00%	Yes	Larva		1	CG
Orthocladiinae							
Brillia sp.	4	0.73%	Yes	Larva		4	SH
Chaetocladius sp.	1	0.18%	No	Pupa		6	CG
Chaetocladius sp.	15	2.73%	Yes	Larva		6	CG
Cricotopus / Orthocladius	3	0.55%	No	Larva	Early Instar	7	SH
Eukiefferiella sp.	3	0.55%	No	Larva	Early Instar	8	CG
Eukiefferiella sp.	6	1.09%	No	Pupa		8	CG
Eukiefferiella Gracei Gr.	6	1.09%	Yes	Larva		8	CG
Hydrobaenus sp.	1	0.18%	Yes	Larva		8	SC
Orthocladiinae	1	0.18%	No	Larva	Early Instar	6	CG
Orthocladius sp.	3	0.55%	No	Pupa		6	CG
Orthocladius sp.	23	4.18%	Yes	Larva		6	CG
Parametriocnemus sp.	1	0.18%	Yes	Larva		5	CG
Stilocladius sp.	1	0.18%	Yes	Larva		3	CG
Tokunagaia sp.	3	0.55%	Yes	Larva		11	CG
Tvetenia Bavarica Gr.	2	0.36%	Yes	Larva		5	CG
Sample Count	550						